

Point-of-Sale (POS) System for Pizza Shop

Software Requirements Specification (SRS)

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1. Introduction

1.1. Purpose

This document is the Software Requirements Specification (SRS) for the Point-of-Sale (POS) System for a pizza shop chain. The purpose of this document is to describe the problem statement, the project objectives, and the stakeholder identification for the portion of the project assigned to the author (Allah Ditta). This SRS follows the Requirement Engineering principles taught in the course material and is written in a clear, testable manner to support later SRS extension and system development.

1.2. Scope

The scope of the assigned deliverable is limited to:

- Problem Statement
- Project Objectives
- Stakeholder Identification

These sections will be integrated into the complete SRS for the full POS system by the project team. The complete system will ultimately provide multi-branch support, dynamic menu management, inventory and cooked item handling, deal recommendations, order processing, reporting, and administrative controls.

1.3. Definitions, Acronyms and Abbreviations

POS Point-of-Sale

SRS Software Requirements Specification

Admin Head-office user who manages branches, menus and reports

Branch Manager On-site manager who manages orders and branch menus

Cashier On-site staff who operate POS terminals

Inventory items Pre-packaged or non-cooked items (e.g., bottled drinks)

Cooked items Items prepared at branch (e.g., pizzas)

Deal A promotional combo or pricing rule that may reduce customer cost

1.4. References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications (informative reference).

2. Problem Statement

Many pizza shops that operate multiple branches face difficulties in managing menus, orders, inventory, and consolidated reporting. Typically, each branch maintains a different menu tailored to local customer preferences. This decentralization often results in:

- Inconsistent menus and prices across branches.
- Difficulty in maintaining and updating branch-specific menus from a central location.
- Errors and delays in sales recording and report generation.
- Poor visibility for administrators over branch-level sales and inventory.
- Inefficient order management leading to preparation delays and customer dissatisfaction.
- Lack of a unified platform for customers to find and use the best available deals automatically.

Therefore, a centralized and smart POS system is required that connects branches, allows branch-specific menus, manages both inventory and cooked items, recommends applicable deals, and produces consolidated reports to support business decision-making.

3. Project Objectives

The objectives below are specific to the tasks assigned to the author and are written to be measurable and verifiable where possible.

1. **Objective 1 (Requirements Documentation):** To produce a clear and professionally formatted SRS containing the Problem Statement, Project Objectives, and Stakeholder Identification for the POS project by November 2025.
2. **Objective 2 (Traceable Scope):** To ensure the documented requirements align with the Requirement Engineering lecture material (including IEEE-style structure) so the document can be integrated into the full project SRS without significant rework.
3. **Objective 3 (Stakeholder Mapping):** To identify and document all key stakeholders, their roles, responsibilities, and interests relevant to the POS system.
4. **Objective 4 (Integration Readiness):** To provide a short integration note and placeholder sections to allow other team members to add Functional, Non-functional, and Use Case sections seamlessly.
5. **Objective 5 (Validation Preparation):** To propose initial validation activities (reviews and prototyping) that the team can use to validate the full SRS.

4. Stakeholder Identification

The stakeholders listed below are identified according to the project scope and the Requirement Engineering slides. For each stakeholder we provide the role, responsibilities, and interests.

Stakeholder	Role / Title	Responsibilities and Interests
Requirement Provider	Client / End-user Representative	Provides requirements, validates functionality, reviews and signs off SRS, and provides feedback during development.
Administrator	System Owner / Head Office	Manages branches, monitors consolidated sales and inventory, creates and updates deals, configures branch policies, and reviews reports.
Branch Manager	Branch-level Supervisor	Operates branch-level management tasks: views and manages orders, updates local menu and availability, monitors branch inventory levels, and resolves customer issues.
Cashier / Staff	POS Terminal User	Processes in-shop orders, records payments, updates order status, and assists in inventory stock updates for sold inventory items.
Customer	End User	Places orders online or in-person, uses deal recommendations, views order status, and provides feedback.
Development Team	Designers and Implementers	Responsible for requirement analysis, SRS preparation, system design, implementation, testing, deployment, and maintenance.
Payment Gateway Provider	External System	Provides card / online payment processing and transaction settlement services.
Printer / Hardware Vendor	External Supplier	Supplies receipt printers, POS tablets, scanners and supports hardware integration and maintenance.

4.1. Stakeholder Communication Plan

- **Requirement Provider / Supervisor:** Weekly meetings and SRS review sessions until sign-off.
- **Administrator / Branch Managers:** Bi-weekly updates and demo sessions for requirements-related clarifications.
- **Development Team:** Daily or frequent coordination via project board (e.g., Trello/GitHub Issues).
- **Customers:** Feedback gathered via beta release and surveys during user acceptance testing.

5. Integration Note and Placeholders

The following placeholders are included to help integrate this document into the full SRS:

5.1. Planned Additional Sections (to be completed by the team)

- Functional Requirements (FR1, FR2, ...): Detailed “The system shall...” statements.
- Non-functional Requirements: Performance, reliability, security, usability statements with quantitative targets where possible.
- Use Case Model: Use case descriptions and diagrams for actors (Admin, Manager, Cashier, Customer).
- System Architecture and Data Model: ERD and component overview.
- Test Cases and Validation Plan: Initial test cases and acceptance criteria.

5.2. Initial Validation Activities

- **Requirements Review:** Formal review of this SRS with supervisor and stakeholders.
- **Prototyping:** Develop a quick interactive POS mockup to validate key workflows (order placement, deal recommendation, menu updates).
- **Test-case Generation:** Create test cases for critical FRs (menu update, order processing, deal recommendations).

6. Approval

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Date: November 2025